

Some FOSS educational software	
Application	Purpose
Euler	Complex numbers and matrices
Kstars	Astronomy with over 1.3 lakh stars, all planets, etc.
Chemlab	Chemistry lab
Sage	Algebra, geometry, etc
Units	Unit conversion
Earth3D	Real-time 3D view of earth
Kalzium	Periodic table and properties of elements
Atomix	Puzzle game for physics
Kig	High precision geometric constructions
Xaos	Fractal geometry

Table 1

include, machine learning with *Weka* (full book available online), O'Reilly publications, *NL Toolkit* (full book on this available, along with the tool kit), the Linux Documentation Project, and so on.

FOSS for education

It is in the area of software resources that the open movement has contributed most to the education sector. The software relevance to education is from three different angles, and these are described briefly in the following sub-sections.

FOSS learning resources

E-learning is another buzzword that is popular among all academic communities, though its meaning and adoption varies widely from group to group. One major concern in e-learning is the quality of content. Traditional content has been largely text and static pictures/images, limited by the medium of textbooks. Much of e-learning content is still restricted to these two.

Animations, simulations and interactive problem solving environments (IPSE) can significantly enhance the teaching learning process. It provides an opportunity to use the multiple senses in absorbing a concept, and also to try out the concept in perhaps restricted environments, through simulation and IPSEs.

These are generally hard to develop, as they involve a significant amount of software development for each topic. The system needs to have a fairly sophisticated model of the content relating to the domain, and be able to recognise and react to the events with respect to the domain. For example, a program illustrating the concept of projectile motion needs to be able to compute the path of the projectile based on relevant parameters—the initial velocity, the angle of throw and gravity. As these parameters are varied by the student, the system needs to revise the computation accordingly. These tools make e-learning much richer than what is possible in a traditional environment, and ought to be part of e-learning.

One reason for the ineffectiveness of e-learning in academic settings is the lack of such quality content, which would deepen the learning and encourage students

to use these. A lot of high-quality programs of this type are available in open source over the Web. Unfortunately, there are no reliable comprehensive repositories for these kinds of programs, as they are scattered efforts from people around the globe.

The UNESCO portal for FOSS, and repositories like the Edubuntu package list, provide some starting places. The OSCAR project of IIT-B also makes an attempt to collect animation programs. IPSEs are not included here, since these are often fairly large programs. Table 1 provides a (very small) sample of resources one can find on the Web.

One major challenge in using these resources for education is the need to link them explicitly into the curriculum. Except the highly-motivated students, most would be lost when exposed to these tools as a collection for them to explore on their own. For the purposeful use of these systems, activities assignments, experiments, etc, need to be formulated, using these tools.

FOSS for basic utilities

This is what's most obvious to people. Today, open source solutions of good quality are available for you to set up a basic computer system, without investing in any proprietary software. All the software components, including the basic operating system, office suite (for documents, presentation, drawing, equations, etc), browser, media players/editors, drawing utilities, network management, and so on, are available in open source today. In most of these cases, one also has a decent number of alternatives to choose from. Table 2 includes some of these tools. Installation and management of these are just as simple or complex as alternative proprietary solutions. FOSS-based desktops are seen to be generally less vulnerable to security problems such as virus infections; this is a major headache for systems administrators in educational establishments, in general.

Full systems customised for the educational sector are also available from some of the popular distributions. Examples are Edubuntu from Ubuntu, Eduknoppix from Knoppix and the proposed EduBoss from BOSS. These include the basic operating system and associated utilities, select tools for educational use (like an equation editor), and some learner resources for specific subjects. This removes the effort of having to pick the relevant packages from various repositories and integrating them individually.

FOSS for learning management

Under this category, I include software that is specifically for the educational environments. There are software solutions for school/college administration, for faculty to run and coordinate the various activities in a course, for faculty to create and manage content for a course, for students to track the progress and collaborate with other students, and so on. Accordingly, here too, the scope of software is vast, and FOSS doesn't disappoint us. Table 2 has a small list of some of these tools.

One can see a wide range of systems, from learning